Analytical Chemistry Cumulative Exam
October 2, 2009

List of Equations.

$$\mu = \frac{1}{2}\sum c_i z_i^{\,2}$$

$$0 = \sum c_i z_i$$

$$\log\gamma = -0.5 z_i^{\,2}\sqrt{\mu}$$

$$\log\gamma = \frac{-0.51 z_i^{\,2}\sqrt{\mu}}{1+(\alpha\sqrt{\mu}/305)}$$

$$[H^+] = \sqrt{\frac{K_1K_2F + K_1K_w}{K_1 + F}}$$

$$\alpha_{HnA} = \frac{[H+]^n}{D}$$

$$\alpha_{H_{n-1}A-} = \frac{K_1[H+]^{n-1}}{D}$$

$$\alpha_{H_{n-j}A} = \frac{K_1K_2\ldots K_j[H+]^{n-j}}{D}$$

$$D = [H+]^n + K_1[H+]^{n-1} + K_1K_2[H+]^{n-2} + \ldots + K_1K_2K_3\ldots K_n$$

A. Warm Up. (20 points)

1. a) Give one example of a buffer system with a pK_a that is temperature dependent.

   b) Explain why it is important to use buffers that have temperature-independent pKa's for microfluidic applications, especially those driven by electrokinetic flows.

2. a) What is the typical expected pH of water from a distilled deionized water unit?

   b) Write the chemical equations describing the equilibrium chemistry that occurs in water sitting on a standard laboratory bench.

B. Real Problems (80 points)

3. Consider two separate 10 mL buffer solutions.
   a) 0.1 F sodium acetate, pH 4.5 (pKa =4.75)
   b) 0.1 F disodium citrate, pH 4.5 (pK_{a1} = 3.128, pK_{a2} = 4.76, pK_{a3} = 6.396)

   a) Calculate the pH of the resulting solution when 1 mL of 1.0 M HCl is added to each buffer system.

   b) Which buffer has the greater buffer capacity?

4. Consider a solution that contains 0.1 M Tris and 0.1 M acetic acid (pK_a = 4.75).
   Tris = Tris(hydroxymethyl) aminomethane), pKa = 8.072
   $(HOCH_2)_3CNH_3^+$ **Abbreviate BH^+**

   Using charge and mass balance correctly, derive (DO NOT SOLVE) an equation that could be used to find the pH of this system.

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